

A Survey on End-to-End Autonomous Driving Training from the Perspectives of Data, Strategy, and Platform

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Abstract—Autonomous driving is a cornerstone technology for the future of intelligent transportation, where end-to-end learning has emerged as a transformative paradigm that directly maps multimodal sensory inputs to driving actions through unified differentiable models. While offering advantages, the effectiveness of end-to-end autonomous driving (E2E-AD) is ultimately determined by the quality of its training ecosystem. This paper provides a comprehensive review of training methods and ecosystem for E2E-AD. We introduce a Data-Strategy-Platform taxonomy that conceptualizes training as an interdependent system. The data layer defines what can be learned, the strategy layer governs how learning aligns with driving objectives, and the platform layer supports scalability and continuous evolution. Within this framework, we survey recent advances across data-centric pipelines, learning paradigms, and training infrastructures, and analyze their interplay in shaping model performance, robustness, and deployability. Finally, we reflect on current limitations and articulate a forward-looking vision that emphasizes a shift from data quantity to data value, from isolated optimization to foundation-driven generalization, and from static training to integrated training-testing loops, aiming toward robust, scalable, and trustworthy autonomous driving systems. We maintain a continuously updated repository tracking cutting-edge literature and works at Our Project Page.

Index Terms—Autonomous Vehicle; End-to-End; Artificial Intelligence; Intelligent Transportation System

I. INTRODUCTION

AUTONOMOUS driving has emerged as a strategic frontier in intelligent transportation systems, offering prospects for safer mobility, higher traffic efficiency, and new paradigms of urban transport [1], [2]. Traditional research has largely followed a modular pipeline architecture, where perception, prediction, planning, and control are separated into distinct subsystems. Although this decomposition facilitates interpretability and engineering controllability, modular systems are prone to error accumulation across interfaces and

struggle to achieve global optimization in highly interactive and mixed traffic environments [3], [4].

In response, end-to-end autonomous driving (E2E-AD) has attracted significant research attention. By mapping multimodal sensory inputs directly to driving actions through unified differentiable models, E2E-AD reduces system coupling and allows joint optimization of perception, decision-making, and control [5]–[7]. Such integration has demonstrated strong potential for handling complex interactions and long-horizon reasoning [8]. Moreover, recent progress in large-scale representation learning, generative models, and foundation models further expands the capabilities of E2E-AD, pushing the research frontier from concept demonstrations toward practical deployment.

Despite these structural advantages, E2E-AD still faces challenges in large-scale real-world application, and its development bottleneck lies not only in model design but also in the training process itself. As models become larger and more capable [9], their effectiveness increasingly depends on whether the training ecosystem can provide: (i) data with sufficient coverage, quality, and governance to capture rare but safety-critical scenarios; (ii) strategies for optimization that can align multi-task objectives and stabilize long-horizon learning; and (iii) platforms with the computational scale and lifecycle management required for continuous model evolution. These three dimensions jointly define the performance boundary of E2E autonomous vehicles.

We clarify the positioning of this review against representative surveys in Table I. Existing studies provide valuable reviews of datasets [10]–[12], perception modules [13], [14], end-to-end architectures [7], [8], imitation learning [6], and foundation models [9], [15], [16]. In contrast, this survey focuses on the training ecosystem and examines how data, learning strategies, and platforms jointly determine training effectiveness.

The main contributions of this survey are as follows:

- A Data-Strategy-Platform taxonomy is introduced, which frames E2E-AD training as a system-level challenge rather than an isolated algorithmic problem.
- A comprehensive review of recent advances is conducted, encompassing data, paradigms, and infrastructures, with particular emphasis on both the current state of the art and the dynamic interplay among these three dimensions.
- Beyond surveying existing work, a future vision for training E2E-AD is proposed, where the shift from data

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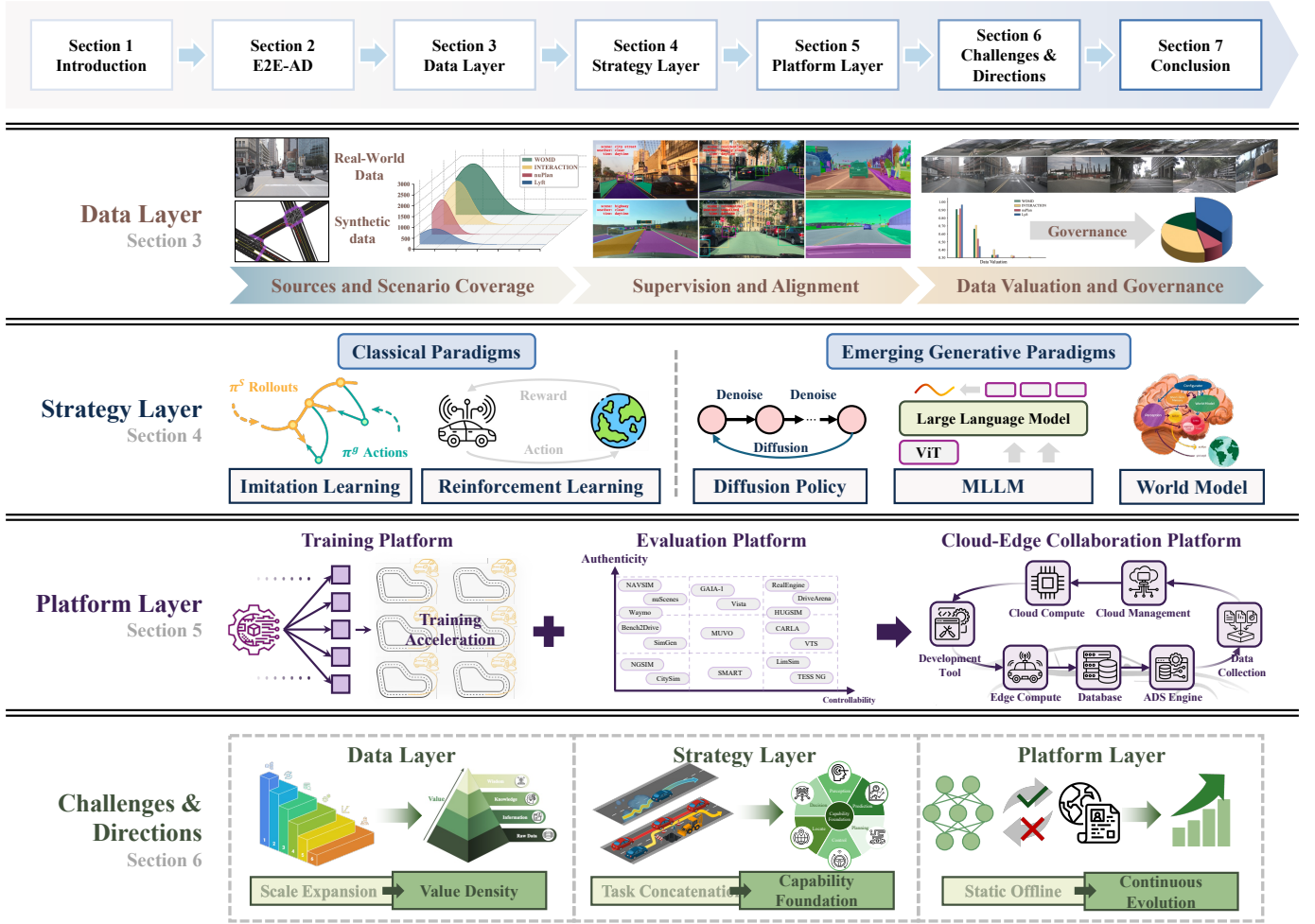


Fig. 1. Overall structure of the survey. The review is organized around a three-layer training ecosystem. (i)*Data Layer* illustrates the progression from raw sources to value-driven governance. (ii)*Strategy Layer* contrasts classical paradigms (IL/RL) with emerging generative models (Diffusion, MLLM, World Models). (iii)*Platform Layer* shows the infrastructure support from training acceleration to cloud-edge collaboration. (iv)*Challenges and directions* highlights the transition toward value density and continuous evolution.

quantity to data value, from task-specific learners to universal foundation models, and from static offline training to continuous training-testing integration is highlighted.

The overall logical structure of the review is shown in Fig. 1. The Data Layer (Section III) establishes the learnable information boundary through scenario coverage and governance, which supports the Strategy Layer (Section IV), where paradigms ranging from imitation learning to world models are reviewed. Finally, the Platform Layer (Section V) provides the computational foundation for training and validation. Section VI discusses the cross-layer evolution shown in Fig. 1.

II. END-TO-END AUTONOMOUS DRIVING

A. Development and Trends

Early autonomous driving systems largely relied on modular architectures, which offer interpretability and engineering modularity but suffer from error propagation and limited global optimization in complex mixed traffic. With the advancement of deep learning and multimodal sensing, E2E-AD weakens information barriers between modules and enables integrated modeling of semantics, interaction, and control by

mapping raw sensory inputs directly to driving actions through unified differentiable models [5]. The earliest demonstrations, such as ALVINN and later NVIDIA’s DAVE-2 [19], [20], showed that neural networks could replace rule-based pipelines with data-driven control policies. While pioneering, these approaches relied on single networks with direct control outputs and struggled to cope with long-tail scenarios or domain transfer.

TABLE I
COMPARISON WITH REPRESENTATIVE SURVEYS

Survey	Data	Str.	Plat.	Main focus
Zhou et al. [15]	✗	✓	✗	MLLMs
Cui et al. [9]	✗	✓	✗	MLLMs
Chen et al. [7]	✓	✓	✗	E2E Architecture
Zhao et al. [17]	✗	✓	✗	Deep Learning
Wu et al. [18]	✗	✓	✓	Reinforcement Learning
Jiang et al. [16]	✗	✓	✗	MLLMs
Wang et al. [11]	✓	✗	✗	Datasets
Liu et al. [10]	✓	✗	✗	Datasets
Ours	✓	✓	✓	Training ecosystem

Subsequent work introduced more structured forms of end-to-end learning, often referred to as modular end-to-end approaches. Instead of discarding intermediate representations, they embedded them within a differentiable framework, allowing perception, prediction, and planning to be optimized under a shared objective. Systems such as MP3, UniAD, and DiffStack [4], [21], [22] exemplify this trend to integrate multimodal information and by incorporating differentiable planning modules to improve consistency between perception and control, which alleviates the interpretability issues associated with integrated networks while retaining the benefits of joint learning.

More recently, the field has increasingly embraced generative end-to-end paradigms that capitalize on advances in language models, diffusion models, and related approaches [9], [15], [23]. These systems move beyond reactive mappings by anticipating long-horizon interactions and synthesizing counterfactual scenarios within latent space. For example, frameworks such as LMDrive and OpenDriveVLA align vision, language, and action in a shared semantic space, thereby facilitating the natural integration of high-level instructions [24], [25]. Meanwhile, world models such as DriveDreamer and GAIA generate scene rollouts that capture rare or cross-domain conditions [26]–[29]. These advances mark a transition from narrowly engineered control policies toward models with enhanced capacities for reasoning, generalization, and even imagination.

Alongside architectural evolution, research trends reveal deeper transformations in how E2E-AD is conceptualized. The data paradigm is moving from scale expansion toward scenario prioritization, with increasing attention to rare, high-risk events and generative supplementation [10], [30], [31]. Training objectives are shifting from perception accuracy to consistency with planning and safety, motivating the exploration of imitation learning, reinforcement learning, and generative models [32]–[34]. Evaluation protocols are evolving from offline metrics to closed-loop benchmarks such as CARLA and nuPlan, which emphasize safety, comfort, and compliance under continuous deployment cycles [35]–[38]. These trends illustrate that the progress of E2E-AD is not merely an architectural evolution but a systematic reconfiguration of training logic.

B. Core Training Challenges

As E2E-AD integrates perception, prediction, planning, and control into a unified framework, its success is determined less by any single algorithmic advance and more by the coherence of the training ecosystem. As illustrated in Fig. 2, three dimensions emerge as decisive. The **data layer** defines the boundaries of what can be learned, yet the long-tail distribution of road events, the scarcity of high-quality annotations, and discrepancies between real and simulated data severely constrain generalization. The **strategy layer** governs how models optimize toward driving objectives, but multi-task conflicts, unstable convergence, and the misalignment between offline proxy metrics and online performance frequently undermine reliability. The **platform layer** provides the engineering

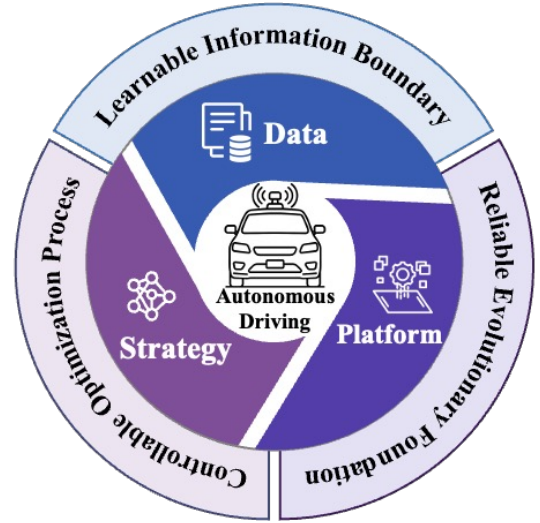


Fig. 2. Interrelationship among data, strategy, and platform layers, which illustrates the systemic coupling of the three layers that constitute the E2E-AD training ecosystem. The data layer defines the learnable information boundary, the strategy layer governs the controllable optimization process, and the platform layer provides the reliable evolutionary foundation, which jointly determine the upper bound of autonomous driving capability.

foundation, where bottlenecks in data pipelines, distributed training, and continuous adaptation often limit scalability and reproducibility.

These challenges are not independent. Insufficient scenario coverage amplifies optimization instability; fragile optimization increases the burden on computational infrastructure; and platform inefficiencies in turn restrict the feasible scope of data and strategies. Well-known issues such as hallucinations, catastrophic forgetting, or cross-domain degradation often manifest when alignment breaks down across these three layers. Thus, the core challenges of E2E-AD are fundamentally systemic, arising from the need to coordinate data, strategies, and platforms into a closed-loop training process. This interdependence motivates the taxonomy adopted in the remainder of this review, which analyzes training methods along the axes of data, strategy, and platform to provide a unified perspective on the development of E2E-AD.

III. DATA LAYER

A. Sources and Scenario Coverage

Scenario coverage defines the learnable boundary of E2E-AD. It is not a matter of accumulating more frames, but of exposing the policy to interactions that shape closed-loop safety, robustness, and generalization. A scenario is therefore treated as a temporally bounded traffic interaction conditioned on road geometry, agents, rules, and environment; its value depends on its expected contribution to planning-relevant performance rather than its frequency alone.

Data sources can be organized into three complementary forms. *Real-world data* from instrumented fleets [39], [40], shadow-mode operations [41], roadside sensing [42]–[46], and crowdsourced vehicles [47], [48] remain the foundation of E2E-AD training for their naturalistic interactions and region-specific driving habits, yet is costly and inherently stochastic,

TABLE II
REPRESENTATIVE DATASETS ON DATA SCALE AND VALUE

Dataset	Year	Scale	Camera	LiDAR	RaDAR	QA	Type	Data Source
(A) Emphasizing Data Scale								
DDD17 [69]	2017	12 hours	✓	✗	✗	✗	Real	Europe
BDD [70]	2017	10000+ km	✓	✓	✗	✗	Real	China
DDD20 [71]	2020	51 hours	✓	✗	✗	✗	Real	USA
A2D2 [72]	2020	41,277 clips	✓	✓	✗	✗	Real	Germany
Nuscenes [51]	2021	40K clips	✓	✓	✓	✓*	Real	USA & Singapore
DriveLM [73]	2023	188K frames	✓	✓	✗	✓	Virtual & Real	CARLA & Nuscenes
DriveMLM [74]	2023	8 maps	✓	✓	✗	✓	Virtual & Synthetic	CARLA
CoVLA [75]	2024	6M frames	✓	✗	✗	✗	Real & Synthetic	Japan
OpenScene [76]	2024	120+ hours	✓	✓	✗	✗	Real	USA & Singapore
PhysicalAI [77]	2025	310,895 clips	✓	✓	✓	✗	Real	Global
(B) Emphasizing Data Value								
Bench2Drive [35]	2024	13,628 clips	✓	✓	✗	✓	Virtual & Synthetic	CARLA
DriveCoT [78]	2024	1,058 scenarios	✓	✓	✗	✓	Virtual & Synthetic	CARLA
StyleDrive [79]	2025	30K scenarios	✓	✓	✗	✓	Real	OpenScene
AoTA [80]	2025	4,800 clips	✓	✗	✗	✗	Synthetic	BDD100K
TAD-E2E [81]	2025	1M frames	✓	✓	✗	✗	Real	Unknown
WOMD-E2E [82]	2025	4,021 clips	✓	✓	✗	✓*	Real	USA

* refers to supplementary data, not the dataset itself.

especially for long-tail and safety-critical events [49]–[54]. In contrast, *Virtual data* from high-fidelity simulators [38], [55], [56], 3D reconstructions [57]–[59], or generative world models [3], [26], [27], [60]–[62] offer controllable expansion of rare or risky conditions but require careful calibration to preserve behavioral realism and prevent distributional drift from real-world dynamics [40], [63]–[65]. Emerging alongside these is *Synthetic data*, produced by generative AI models and typically released after being manually verified.

In recent years, world models are reshaping the data layer. They learn generative dynamics that can produce temporally coherent, action- or trajectory-conditioned futures. CosmosDrive-Dreams, DriveDreamer and GAIA show how such models can synthesize driving scenes, replay counterfactual futures, and connect visual generation with motion-level objectives [26]–[29], [66]. Thus, when used for scenario generation, long-tail augmentation, failure replay, or virtual-real fusion, world models act as data engines; when reused for planning or policy optimization, they enter the Strategy Layer.

Effective coverage therefore requires a value-guided hybrid acquisition strategy. Real-world data provides distributional grounding; simulation offers rule-controllable perturbation; and curated synthetic datasets deliver quality-controlled samples. [67], [68] As the field transitions from quantitative accumulation to qualitative optimization, the landscape of available datasets has shifted accordingly. Table II categorizes representative benchmarks, distinguishing between earlier datasets emphasizing sheer scale (Part A) and recent efforts prioritizing scenario value and rarity (Part B).

B. Supervision and Alignment of Labels

Supervision transforms raw data into learnable signals, enabling E2E-AD systems to distinguish between behaviors that should be learned and those that should be avoided. As raw sensor data lacks semantic or behavioral grounding, it is annotation that serves as the mechanism for assigning value and meaning to each scenario within the operational design domain (ODD) [10]. Therefore, the goal of supervision is not simply to label data exhaustively, but to produce training signals that are consistent with scenario value and aligned with the system’s planning and safety objectives.

Manual annotation provides high-fidelity supervision by capturing precise object semantics, trajectories, and interaction cues [83], [84]. However, it is inherently costly and difficult to scale to the volume required by E2E-AD. Automatic annotation based on pre-trained models, cross-modal fusion, or pseudo-labeling [85]–[88] offers scalability but introduces risks of misalignment, temporal drift, and noise [7], [10]. In practice, an effective annotation strategy should combine manual annotation for high-value scenarios that are safety-critical or rare and automated or semi-supervised approaches for low-value, high-frequency scenarios where approximate supervision is sufficient.

However, not all labeled data contribute equally to strategy improvement. Unlike previous approaches that equated long-tail scenarios with high-value scenarios, this paper argues that scenarios with higher value can bring greater performance gains, especially those that expose decision ambiguities, complex interactions, or boundary conditions of the driving policy, as depicted in Fig. 3. As a result, annotation priorities should follow a value-guided hierarchy, with detailed and multimodally aligned labeling for high-value scenarios, while

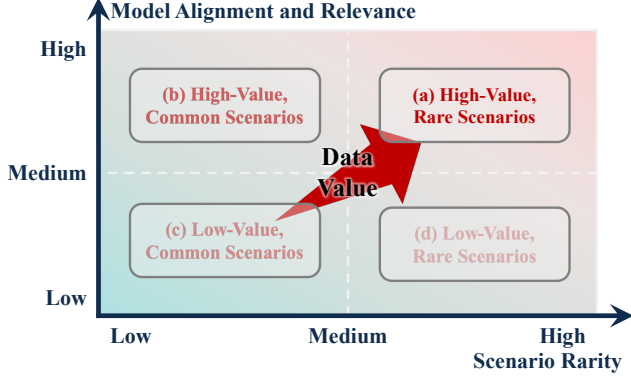


Fig. 3. Illustration of classification of scenarios by data value, which divides scenarios into four categories, highlighting the importance of prioritizing high-value scenarios for efficient training.

efficient and lower-fidelity supervision for common, low-impact ones. This approach ensures that annotation resources are concentrated where they most enhance planning stability and safety metrics, rather than where labeling is easiest or most frequent.

C. Data Valuation and Governance

As the control system of the data layer, data valuation and governance aim to allocate resources toward the most informative and safety-relevant scenarios, thereby accelerating model convergence and improving overall training efficiency [89]–[91]. While early industrial practices often pursued data accumulation through sheer volume [77], the complexity and long-tail characteristics of real-world driving environments suggest that massive low-value samples yield rapidly diminishing marginal returns. Consequently, the paradigm must shift from scale expansion to value density, where the training benefit is determined by the quality of scenario coverage rather than total sample count.

To implement this shift, a quantifiable scenario value system is essential. It summarizes recurring ideas in scenario selection, uncertainty-based mining, and data governance into a simple value-scoring view. We define the value of a scenario $V(s)$ not as a static property, but as its marginal contribution to the model’s current capability boundary [92]. In practice, this can be operationalized through a composite scoring function that integrates risk, uncertainty, and rarity:

$$V(s) = w_r \cdot R(s) + w_u \cdot U(s|\theta) + w_d \cdot D(s|\mathcal{D}_{train}) \quad (1)$$

where $R(s)$ quantifies safety-criticality based on physical metrics, $U(s|\theta)$ represents the model’s epistemic uncertainty regarding the scenario and $D(s|\mathcal{D}_{train})$ Measures the distributional distance of scenario s from the existing training distribution $\mathcal{D}_{train}$.

Critically, scenario value is dynamic; as the model θ evolves, previously difficult scenarios ($U(s|\theta) \downarrow$) lose their training utility, while new edge cases gain importance [62], [93]. Effective governance requires a data engine that continuously re-evaluates samples. This involves establishing version trees

that map specific data subsets to model checkpoints, ensuring that every performance improvement or degradation can be traced back to the addition or removal of specific high-value scenarios [94], [95]. By rigorously managing data lineage and prioritizing resources based on the $V(s)$ metric, the training ecosystem transforms from a passive repository into an active optimization engine, ensuring that computational resources are concentrated where they yield the greatest robustness benefits.

IV. STRATEGY LAYER

The E2E-AD strategy layer maps multimodal perception to control actions, governing closed-loop stability in mixed traffic. To transcend the efficiency-robustness trade-offs inherent in Imitation Learning (IL) and Reinforcement Learning (RL), the field is pivoting toward generative paradigms. Approaches such as diffusion policies, multimodal language models, and world models introduce advanced reasoning and adaptability capabilities. As shown in Table III, these paradigms possess distinct, complementary advantages.

A. Imitation Learning

IL trains an E2E-AD policy directly from expert demonstrations, avoiding explicit reward engineering and yielding strong data efficiency at initialization [96]. Canonical instances include behavior cloning (BC) [97]–[99] and distribution-matching variants such as inverse reinforcement learning (IRL) [100], [101] and generative adversarial imitation learning (GAIL) [102], [103], which collectively demonstrate that policy-level knowledge can be distilled from human drivers.

Despite their practicality, IL methods face three structural limitations in autonomous driving. First, they exhibit a high dependence on expert data, where demonstrations are costly to collect, often noisy, and may contain suboptimal behaviors [6], [104]. Second, they suffer from covariate shift, as models trained on expert trajectories may encounter unseen states at deployment [105], [106], leading to compounding errors and potential loss of control. Third, real-world driving data follow a long-tail distribution, where critical but rare events are under-represented, leaving IL-trained policies fragile under safety-critical conditions [107], [108]. These challenges constrain the reliability and generalization of IL in real deployment.

To address these limitations, substantial effort has been devoted to optimizing IL training. On the data side, augmentation strategies such as viewpoint transformation, trajectory perturbation, and synthetic counterfactual construction enrich the training distribution and mitigate overfitting to narrow expert trajectories [109]–[113]. On the algorithmic side, dataset aggregation (DAgger) introduces iterative corrections by allowing the agent to query experts in newly visited states [114], while SafeDAgger further reduces expert intervention costs by activating corrective supervision only in high-risk conditions [115]–[117]. Adversarial extensions, such as GAIL and its robust variants, improve generalization by aligning trajectory distributions rather than replicating individual actions [103], [118]. More recently, IL has been integrated with generative models. World models can simulate imagination rollouts of rare or unseen states, providing synthetic replay data to reduce

TABLE III
COMPARISON OF FIVE E2E-AD TRAINING PARADIGMS

Paradigm	Multi modal	AR error	Traj-level	RT ready	Long-horizon	Low intera.	Interp.	Best role	Bottleneck
IL	✗	✗	✗	✓	✗	✓	✗	Fast policy prior; Supervised initialization	Closed-loop drift; Long-tail blind spots
RL	✗	✗	✗	✓	✓	✗	✗	Objective-aligned refinement; Closed-loop optimization	Reward and exploration risk; Sim-to-real gap
DP	✓	✗	✓	✗	✗	✓	✗	Multimodal trajectory proposer; Timing consistency	Sampling latency; Hard-constraint guarantees
MLLM	✓	✓	✗	✗	✓	✓	✓	Semantic reasoning; Intent inference	Hallucination and control gap; Real-time cost
WM	✓	✓	✓	✗	✓	✓	✗	Imagination-based planning; Closed-loop generation	Rollout drift; Model bias and hallucination

AR error = autoregressive error accumulation; Traj-level = generates trajectories as coherent objects; RT ready = natively compatible with tight closed-loop latency; Low intera. = does not require large-scale on-policy interaction.

distribution shift without incurring additional safety risks during online collection [119], [120]. Together, these techniques transform IL from a purely passive paradigm into an adaptive training mechanism with improved robustness and coverage.

In summary, IL is better viewed as a *policy bootstrapper* rather than a final alignment mechanism in the training ecosystem, which provides a strong prior over normal driving, but isn't able to enforce long-horizon safety under distribution shift by itself. In E2E-AD applications, IL is more suitable for integration with RL for safety alignment and with generative paradigms for scenario imagination and rare-event robustness.

B. Reinforcement Learning

RL optimizes a driving policy through interaction, improving long-horizon returns under an environment-defined feedback signal (MDP/POMDP) [121]–[123]. In E2E-AD, RL's distinctive value reflects on refining a policy toward long-horizon objectives (safety margins, comfort, compliance) that are difficult to fully encode in supervised labels rather than learning to drive from scratch [18], [124].

However, the direct application of RL to E2E-AD encounters four systemic challenges. First, sample inefficiency remains a central bottleneck, as convergent policies often require millions of interactions, while real-world data collection is costly and unsafe [90], [125], [126]. Second, safe exploration is particularly problematic in driving, since unconstrained trial-and-error may lead to collisions, boundary violations, or traffic infractions, which are unacceptable in deployment [127], [128]. Third, reward misalignment arises because real driving objectives are multi-dimensional which covers safety, comfort, efficiency, compliance and so on, but are often compressed into scalar reward signals, producing biased optimization and divergence between offline metrics and online safety [129], [130]. Finally, cross-domain generalization is difficult because policies trained in one environment may degrade severely when transferred across cities, weather, or traffic cultures, while sim-to-real gaps further amplify the distribution shift [131], [132].

To mitigate these issues, diverse optimization strategies have been proposed. For sample efficiency, offline RL leverages large-scale replay datasets to learn policies without costly

online exploration, often regularized to prevent extrapolation errors on out-of-distribution actions [133]. Hybrid approaches combine imitation learning with reinforcement learning, where demonstrations provide fast initialization and RL subsequently fine-tunes policies through interaction [134]–[136]. World models further enhance efficiency by enabling imagination-based rollouts in latent space, thereby generating synthetic trajectories that extend coverage of rare or dangerous scenarios [137]–[140]. Regarding safety, constrained RL incorporates explicit constraints such as collision avoidance and lane keeping into the optimization process, while risk-sensitive methods based on Conditional Value at Risk prioritize robustness under tail events [141]–[144]. Shielding techniques, such as control barrier functions or model predictive control filters, can override unsafe actions during execution, reducing risk without halting policy improvement [145]–[147]. For reward alignment, preference-based methods such as RLHF align training signals with human judgments of safe and natural driving behaviors, while hierarchical RL decomposes tasks into high-level decision planning and low-level control [90], [148]–[150], improving interpretability and multi-objective balance. Finally, generalization is strengthened by domain randomization and style perturbation in simulation, continual learning to preserve knowledge across deployments, and curriculum learning strategies that progressively expose policies to increasingly complex scenarios [151]–[153].

As a result, RL is best positioned as a *policy alignment* and *refinement engine* in the training ecosystem for its ability to enforce long-horizon objectives beyond labels. Therefore, RL is often used in conjunction with world models or to fine-tune other paradigms such as IL and LLM. If used directly for E2E-AD, RL requires high-fidelity evaluation environments and guardrails that bound exploration and reward misspecification.

C. Emerging Generative Paradigms

Beyond the classical paradigms of IL and RL, recent advances in generative modeling have opened new avenues for E2E-AD. Unlike traditional methods that directly optimize policies over observed data or interactive feedback, generative approaches aim to model richer distributions of actions, semantics, and environments [3], thereby equipping

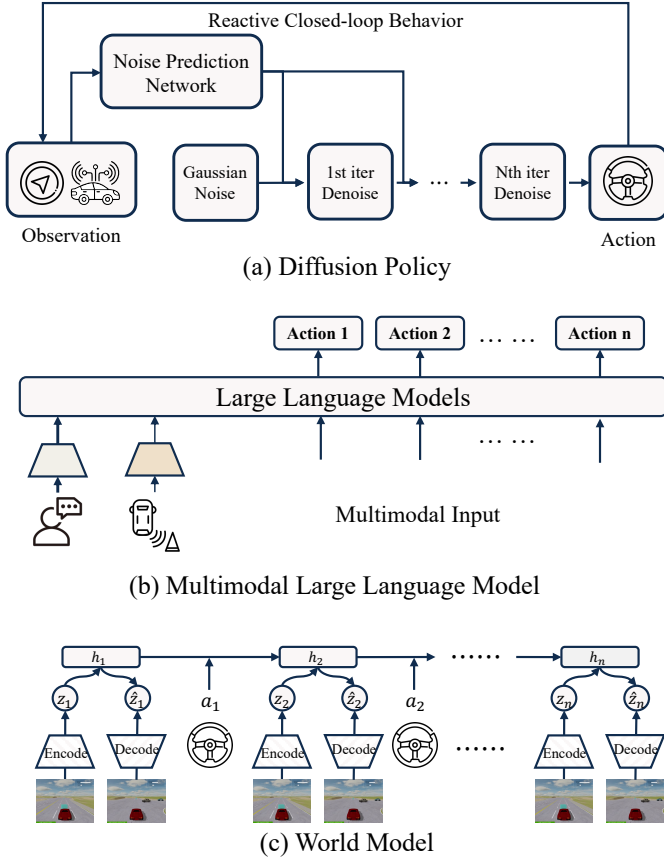


Fig. 4. Structure of Emerging Generative Paradigms for Autonomous Driving. This diagram illustrates the integration of generative models in end-to-end autonomous driving systems, covering the application of diffusion policy, multimodal large language model and world model.

driving agents with capabilities for reasoning, imagination, and adaptation, which signify a transition from purely reactive policy learning toward generative and anticipatory strategies, expanding the scope of training from replication and optimization to reasoning, simulation, and scalable adaptation.

1) **Diffusion Policies:** Diffusion-based policies represent a generative paradigm for end-to-end autonomous driving, modeling *multimodal* action distributions via a denoising process that refines random noise into structured trajectories or control sequences [154], [155]. Unlike deterministic mappings in classical policy learning, diffusion policies inherently capture the stochasticity of driving behavior, allowing policies to represent multiple feasible actions under uncertainty [156]–[158]. This property makes them particularly suitable for highly interactive mixed traffic, where safe and robust behavior often depends on anticipating diverse outcomes rather than committing to a single trajectory.

The strengths of diffusion policies lie in their robustness and expressivity. By modeling the full action distribution rather than a single deterministic path, these methods can naturally account for uncertainty, generate diverse yet plausible driving maneuvers, and avoid mode collapse in decision-making [159], [160]. Such diversity provides resilience in ambiguous or multi-agent interactions, where multiple safe behaviors may coexist. Moreover, the probabilistic nature of diffusion models

facilitates uncertainty-aware planning, offering a principled mechanism to balance risk and exploration in safety-critical driving scenarios.

Despite these advantages, diffusion policies also face several critical challenges that recent research is working to address. The iterative denoising procedure is computationally expensive, elongating training and increasing resource demand [161]. Correspondingly, efficiency-oriented diffusion variants aim to reduce sampling steps and amortize generation costs. Real-time deployment is further challenged by multi-step sampling latency [155], motivating guided or controlled generation that concentrates probability mass on safety-compliant regions and reduces wasted sampling [162]–[164]. Finally, while diffusion is powerful for proposing trajectories, aligning generated outputs with planning objectives and vehicle dynamics is non-trivial, raising concerns about interpretability and integration with safety-critical controllers [165]–[167]. Scenario-conditioned diffusion and planner-oriented adaptations address this gap by injecting structured context and optimizing generations to satisfy kinematic constraints and downstream safety metrics [168]–[172].

In the training ecosystem, diffusion policies are best positioned as a *multimodal proposal and planning component* rather than a standalone E2E controller. They are most applicable when interaction uncertainty is high and multiple safe futures must be explicitly represented, but they remain constrained by sampling latency, controllability under hard constraints, and the difficulty of certifying safety from a generative distribution. In practice, these properties make diffusion particularly well-suited to the role of a trajectory-level planner that converts high-level decisions into diverse and feasible candidates. Moreover, diffusion policies are often most effective in hybrid training pipelines, serving to preserve multimodality and uncertainty awareness in other paradigms.

2) **Multimodal Large Language Models:** Multimodal large language models (MLLMs), including vision-language-model (VLM) and vision-language-action (VLA) systems, represent an emerging direction for training E2E-AD policies, whose underlying principle is to leverage the grounding capabilities of large-scale language and vision-language models to connect raw perception inputs with natural language instructions and low-level vehicle actions [15], [16], [173]. By aligning sensory observations with symbolic or linguistic abstractions, these models aim to bridge high-level reasoning with control execution, thereby enabling autonomous vehicles to understand and respond not only to the physical environment but also to human commands expressed in natural language.

The key strengths of MLLMs lie in their generalizability and interpretability [174]–[176]. Owing to large-scale pretraining across diverse multimodal corpora, these models exhibit strong transferability to new scenarios with minimal task-specific data, a property especially valuable for cross-domain driving deployment [25], [177], [178]. Furthermore, their ability to process natural language enables command reasoning, which allows human operators to guide vehicle behavior intuitively. The integration of linguistic representations also provides a degree of transparency and interpretability absent in purely black-box policies, while their modular pretraining structure

supports flexible adaptation to varied downstream driving tasks [179], [180].

Nonetheless, significant challenges constrain their practical deployment in autonomous driving. First, maintaining temporal consistency is difficult, as LLM-driven policies must integrate sequential sensor data and reason over long horizons, whereas most pretrained models are optimized for static image-text pairs [168], [181]. Second, low-level grounding remains limited, as translating high-level linguistic reasoning into precise, real-time control commands requires robust alignment across abstraction layers [182], [183]. Third, safety and reliability are unresolved concerns, since language models may hallucinate, misinterpret ambiguous instructions, or generate unsafe control suggestions [184]–[186]. Finally, issues of real-time performance pose engineering hurdles, as the inference latency of large multimodal architectures is often incompatible with the stringent time constraints of closed-loop driving [187]–[190].

To mitigate these limitations, researchers have proposed specialized training methodologies. Supervised pretraining on paired perception-language-action datasets improves grounding between natural language and driving trajectories [75], [85], [191], [192]. Instruction tuning further enhances the ability to follow human commands in structured traffic contexts [33], [193], [194]. Beyond supervised alignment, RLHF and its variants provide preference-driven optimization, allowing policies to align with human judgments of safety, comfort, and naturalistic behavior [195]. More recently, chain-of-thought and visual chain-of-thought reasoning have been introduced to enhance interpretability and robustness, by decomposing multimodal inputs into explicit intermediate steps that link perception, linguistic reasoning, and action selection [196]–[198]. This not only improves transparency in decision-making but also reduces error propagation by anchoring reasoning in visually grounded cues.

In the training ecosystem, MLLMs are best interpreted as *semantic decision and coordination modules*, which are effective at mapping complex perception into structured intents, rules, and explanations. However, due to temporal inconsistency and the risk of hallucinated outputs, deploying MLLMs directly without safeguards remains difficult. Practically, MLLMs complement other paradigms by providing verifiable intermediate representations that condition downstream planners and controllers, such as diffusion planners or IL/RL policies.

3) **World Models:** World models (WMs) represent a generative paradigm that aims to endow E2E-AD systems with the ability to learn compact latent representations of environment dynamics and to use these representations for “imagination”-based planning [199]–[201]. Instead of directly mapping sensory observations to actions, world models first construct a predictive latent space that encodes the spatiotemporal evolution of the driving environment. Policies can then be optimized within this latent space by simulating counterfactual trajectories or multi-agent interactions, thereby reducing reliance on costly real-world sampling [202]–[204]. This principle fundamentally shifts the training process from purely reactive behavior learning toward proactive reasoning over possible

futures.

The primary strengths of world models lie in sample efficiency and scenario generalization. By replacing expensive environment interactions with latent rollouts, world models dramatically reduce the need for real-world data collection while still supporting long-horizon policy optimization [205], [206]. Furthermore, their capacity to generate counterfactual scenarios allows the model to explore safety-critical events that may be scarce in naturalistic datasets but crucial for robust deployment [207]. As a result, world models are well-suited for addressing the long-tail problem in E2E-AD.

Nevertheless, significant challenges hinder their practical application. A central issue is the fidelity of generated rollouts, as imperfect latent predictions can diverge from real-world dynamics, undermining policy reliability. Hallucination risks, where the model produces implausible or inconsistent environment states, pose safety concerns when imagination outputs are used for policy training [208]–[211]. Moreover, ensuring stability over long horizons remains difficult, as small errors in latent prediction often accumulate, leading to compounding deviations that degrade policy performance [212].

Recent advances in training strategies seek to address these challenges. Predictive modeling methods focus on learning latent state transitions that preserve physical plausibility and semantic consistency, often through self-supervised or multi-task objectives [26], [60], [213]. Imagination-augmented reinforcement and imitation learning integrate world model rollouts into standard policy optimization, expanding the range of training scenarios without requiring direct real-world exploration [139], [214], [215]. Finally, closed-loop adaptation connects imagined experiences with real-world feedback, continuously refining the latent dynamics model to align imagination with observed behavior. These approaches collectively aim to transform world models from auxiliary simulators into central engines of policy training [216], [217].

In the training ecosystem, world models are best positioned as *imagination and evaluation backbones*. As clarified in Section III-A, their first role is to serve as data engines that generate, replay, and diversify rare or high-value scenarios. Recent works further show that unifying visual scene generation with motion representation can make world-model representations transferable from generation to planning [218]–[222]. However, their effectiveness is ultimately bounded by fidelity, especially with respect to physical rules, causal interaction, and motion consistency rather than visual appearance alone. However, their effectiveness is ultimately bounded by fidelity, especially in terms of physical rules rather than optical visual. It can serve as a joint optimization tool for other paradigms, provide more valuable data for IL, LLM, and other paradigms, and provide closed-loop simulation environment for RL training or evaluation.

V. PLATFORM LAYER

A. Distributed and Parallel Training Platform

For large-scale end-to-end models, the memory and throughput requirements of training exceed the capacity of a single computing device. Consequently, distributed and

parallel training has become a central issue in the infrastructure layer. The design objectives of such systems can be summarized in three dimensions: first, scalability, maintaining near-linear throughput improvement as device scale grows; second, efficiency, achieving efficient overlap and load balancing between communication and computation; third, stability and reproducibility, ensuring numerical consistency and process traceability during long-cycle training. Around these objectives, five complementary parallel paradigms have been developed, which are often combined in hybrid forms in engineering practice.

1) **Data Parallelism (DP)**: DP is able to replicate full models across devices, partition training data into separate forward and backward computation, and synchronize gradients via all-reduce [223]–[225]. It offers ease of implementation, insensitivity to model architecture, and near-linear throughput scaling. However, its drawbacks include high memory overhead for the reason that parameters and optimizer states must be stored on each device and sensitivity to network bandwidth during gradient synchronization. Gradient accumulation, compression, and delayed synchronization are usually used to alleviate bandwidth pressure in engineering practices [226], [227].

2) **Tensor Parallelism (TP)**: TP partitions models at the operator or weight level, distributing large-scale matrix multiplications and convolutions across devices [228]. Each device stores only parameter subsets, reducing memory pressure and enabling training at ultra-large parameter scales, whose main cost is fine-grained cross-device communication, where mismatched partitioning and topology can amplify overhead. Efficient implementation requires fusion and scheduling at the operator level to maximize overlap with computation.

3) **Pipeline Parallelism (PP)**: PP divides models into stages by network depth, enabling micro-batches to flow through the pipeline across stages, which can reduce peak memory via staged storage and checkpointing and improve utilization in deep networks [229]. Its effectiveness, however, is constrained by idle ‘bubble’ time and load imbalance, which can be alleviated by finer-grained stage partitioning, careful tuning of micro-batch scheduling, and applying recomputation at stage boundaries to further reduce activation storage [230], [231].

4) **Sequence Parallelism (SP)**: SP distributes activations and contexts along time or token dimensions, distributing memory and communication costs of long sequence training across devices [232]. It is especially effective for Transformers, extending trainable context length while reducing long sequence memory peaks. Typical implementations, such as DeepSpeed Ulysses [233], use all-to-all communication to partition attention computation. SP integrates well with DP, TP, and PP, though boundary exchanges introduce additional communication. For the excellent performance optimization, the DeepSpeed architecture is widely used for most existing model pre-training or fine-tuning [234]–[237].

5) **Expert Parallelism (EP)**: Mixture-of-experts models expand effective parameter scale without proportionally increasing memory and computation by introducing multiple experts in feedforward layers and performing sparse routing

TABLE IV
COMPARISON OF COMPUTATIONAL REQUIREMENTS FOR
REPRESENTATIVE E2E-AD

Method	Year	Paradigm	Training Hardware*
InterFuser [241]	CoRL 2022	IL	8×V100
UniAD [4]	CVPR 2023	IL	16×A100
DiffusionDrive [162]	CVPR 2025	DP	8×RTX4090
AutoVLA [195]	NeurIPS 2025	MLLM	8×L40
ORION [183]	ICCV 2025	MLLM	32×A800
FSDrive [196]	NeurIPS 2025	MLLM	32×A6000
RS2E [242]	TPAMI 2026	IL+RL +DP	8×A100

* reported experimental hardware, not minimum requirements.

on input tokens, activating only a few experts for computation. Expert parallelism distributes different experts across multiple devices, routing tokens to corresponding experts in all-to-all fashion during the forward pass, then aggregating after computation. Representative systems like Google’s GShard [238] replace some layers in models with MoE layers, distribute different experts across different devices, and use routers for cross-device communication, achieving parameter sparsification in ultra-large-scale models. Microsoft’s DeepSpeed MoE [239], based on the DeepSpeed architecture and combining ZeRO optimizer [240] with All-to-All communication optimization, significantly reduces memory consumption, demonstrating feasible paths for large-scale sparse models and significantly reducing communication and latency costs during training and inference.

6) **Hybrid Parallelism and Strategy Selection**: In practice, employing hybrid parallel strategies is essential for training large-scale E2E-AD models. While DP and TP typically form the foundation for parameter scaling, PP and SP are increasingly critical for optimizing deep architectures and long-context multimodal streams. Additionally, EP offers sparsification-based scaling to expand model capacity within feasible budgets. The essence of these distributed frameworks lies in communication-aware partitioning that balances memory, computation, and bandwidth pressure across the cluster.

This evolution in parallel strategies is directly driven by the escalating hardware demands of modern algorithms. As detailed in Table IV, there is a clear trend toward exponential resource consumption, which is transitioning from classical Imitation Learning models, which were manageable on single-node configurations, to Generative paradigms that necessitate massive multi-node clusters to handle immense parameter counts. This non-linear growth not only validates the necessity for advanced hybrid parallelism but also highlights a widening compute gap between training infrastructure and energy-constrained vehicle edge devices. Consequently, future platforms must extend beyond scalable training to include robust pipelines for model distillation and quantization, ensuring that large-scale foundation models remain deployable in real-world environments.

B. Testing and Evaluation Platform

As an essential component of the E2E-AD training system, the testing platform is also responsible for the performance, safety, and reliability of models in real-world or simulated environments [243]–[245].

1) **Simulation-Based Testing and Closed-Loop Validation:** Simulation-based testing plays a pivotal role in evaluating E2E-AD models in a controlled, repeatable environment. High-fidelity simulators such as CARLA, LGSVL, and NVIDIA DRIVE Sim allow models to be tested across a wide range of predefined scenarios, including edge cases and rare occurrences that would be costly or dangerous to replicate in real-world trials [35], [38], [246]–[249]. Publicly available descriptions indicate that Waymo emphasizes simulation-first validation, using its simulation platform [250] to replicate billions of virtual miles prior to limited on-road deployment. These platforms provide virtual environments that can accurately model road conditions, traffic patterns, and weather variations, ensuring comprehensive coverage of possible driving situations. However, simulation-to-real transfer remains a critical challenge because differences between simulated and real-world environments can lead to discrepancies in model performance, especially when dealing with rare events [251], [252].

2) **Real-World Testing and Real-Time Evaluation:** While simulation provides valuable insights, real-world testing is indispensable for validating E2E-AD systems under actual driving conditions [253], [254]. On-road testing exposes models to dynamic, unpredictable environments, including complex human behaviors, varying traffic conditions, and environmental factors such as weather or road surface variations [255]. The challenge in real-world testing is to ensure safe operation of the vehicle while evaluating the model's adaptability to real-time data streams and decision-making. Closed-loop real-world testing integrates real-time sensor data with the model's decision-making process, allowing for feedback on how well the model adapts to unforeseen situations. A robust real-world testing platform must incorporate monitoring and safety mechanisms to mitigate risks associated with algorithm instability, though these safeguards may sometimes mask the model's true performance. Therefore, hybrid virtual-real testing platforms may offer a promising solution for balancing safety with accurate performance evaluation [252], [256]–[258].

Based on publicly available descriptions, industrial E2E-AD platforms appear to differ along two broad tendencies, although the internal details of proprietary systems are not fully observable. A verification-driven, simulation-centric regime, represented by Waymo, treats safety as a pre-deployment property that must be demonstrated through repeatable, scenario-addressable evidence; accordingly, it prioritizes high-fidelity simulators, rule-based scenario generation, and deterministic regression pipelines that support reproducibility, interpretability, and regulatory traceability. In contrast, an experience-driven, fleet-centric regime, represented by Tesla, treats safety and generalization as emergent properties that improve through large-scale exposure to real operating conditions; it therefore prioritizes event-triggered data capture, rapid data-to-

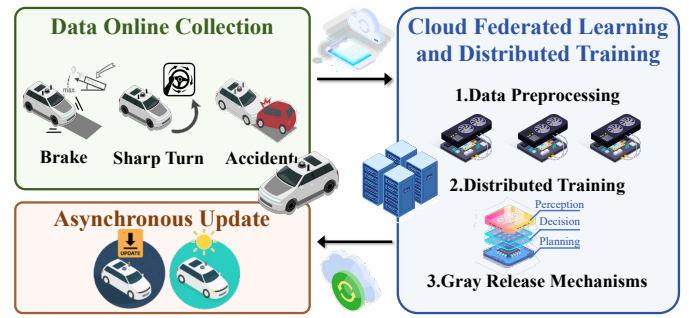


Fig. 5. Continuous autonomous driving training framework based on end-cloud collaboration, which illustrates the closed-loop process of continuous model evolution. Vehicle-side data collection captures high-value driving events, which are uploaded for cloud-based federated and distributed training. Through asynchronous updates, models are incrementally improved and redeployed, achieving sustainable and adaptive evolution.

training conversion, and continuous deployment mechanisms that shorten the discovery-to-fix cycle for long-tail failures [259]. These two concepts are not mutually exclusive, and emerging companies such as XPeng and Huawei may combine elements of both.

3) **Testing Metrics and Performance Validation:** The efficacy of testing platforms is fundamentally determined by the evaluation metrics used to measure the model's performance [35], [36]. Traditional metrics, such as accuracy in object detection or classification, are insufficient to fully capture the complexity of autonomous driving tasks [260]. Instead, testing platforms must incorporate a broader range of metrics that assess safety, comfort, and efficiency. Key performance indicators include collision rates, the ability to perform emergency maneuvers, and compliance with traffic laws. Additionally, metrics such as acceleration smoothness that relate to user comfort are also critical in evaluating the model's effectiveness in providing a human-friendly driving experience [261]–[264].

The main challenges in testing platforms revolve around scalability, realism, and safety. As E2E-AD models grow in complexity, testing platforms must scale accordingly, handling increasing amounts of data and higher computational requirements. However, most existing tests discover problems through massive testing, which often wastes excessive resources. How to accelerate testing and training by leveraging concepts such as scenario value may be an important research topic for the future.

C. Cloud-Edge Collaboration and Continuous Training Platform

As autonomous driving systems move from research and development to actual deployment, models face increasingly complex environmental dynamics and regional differences [265]. Relying solely on offline training can no longer meet long-term stability and cross-domain adaptability requirements. Public information suggests that XPeng and Huawei ADS [266] have explored cloud-edge collaboration frameworks related to Fig. 5, including event-driven collection, staged updates, and cloud-supported training pipelines. Cloud-edge collaboration and continuous training thus become key

paths for improving system evolution capability [37], [267]–[269]. Their core objective is to establish closed-loop pathways between vehicle-side and cloud-side: the vehicle side focuses on event-triggered data collection and lightweight processing, while the cloud side provides large-scale computing support, enabling asynchronous incremental model updates and cross-regional adaptation.

Thus, cloud-edge collaboration not only achieves decoupling and coordination between training and deployment but also provides dynamic adaptation and continuous optimization capabilities for autonomous driving models, serving as core technical guarantees for system large-scale deployment.

1) **Online Collection and Edge Preprocessing:** After deployment, vehicle-side systems should use event-driven collection strategies as core, prioritizing the recording of samples with high marginal value for safety and robustness, such as emergency braking, sharp turns, close encounters, abnormal yielding, and perception divergence. Vehicle sides complete preliminary quality filtering and redundancy removal, including time synchronization correction, basic calibration verification, and compression encoding, performing lightweight anonymization and desensitization processing locally as much as possible, then uploading in grades according to bandwidth and priority. Compared to traditional offline collection, this mechanism continuously supplements long-tail scenarios at low additional cost and provides high-frequency fresh samples during regional migration and seasonal changes.

2) **Asynchronous Updates and Model Governance:** Continuous training needs to break hard constraints of synchronous cycles, forming asynchronous mechanisms where data flow, training flow, and deployment flow advance in parallel. After data enters training platforms, incremental training jobs are automatically triggered, producing candidate models and registering them in model repositories. Each candidate version must be bound to clear data versions, configurations, and checkpoints, completing offline metrics and replay set evaluation before entering online validation. To shorten improvement realization time, stable job orchestration and resource scheduling strategies should be established, enabling deep overlap of decoding, augmentation, communication, and computation to improve throughput and timeliness. Meanwhile, simple and clear promotion criteria should be set to ensure update rhythms are predictable and results interpretable.

3) **Gray Update and Rollback Mechanisms:** New models should not go fully online at once. Gray release follows principles of small scale and controllable scope, first enabling new versions in limited fleets or specific regions, continuously observing safety-related metrics, interaction quality metrics, and system loads, then gradually expanding coverage in phases. To prevent occasional mismatches from evolving into systematic risks, one-click rollback channels must be established: when degradation alerts are triggered or preset thresholds are exceeded, immediately switch to the previous stable version and automatically archive related data to support subsequent localization and review. The coordination of gray release and rollback makes trial and error controllable, enables rapid damage control for failures, while providing high-value failure samples for continuous training.

Industrial efforts toward scalable testing and continuous improvement have increasingly shaped the development of E2E-AD. Publicly reported practices from companies such as Waymo, Tesla, Cruise, Baidu Apollo, XPeng, and Huawei suggest different philosophies regarding generalization, safety validation, and update efficiency. To illustrate this landscape, Table V summarizes representative approaches based on public sources; the entries should be read as indicative patterns rather than exhaustive descriptions of proprietary systems.

VI. OPEN CHALLENGES AND FUTURE DIRECTIONS

E2E-AD is currently transitioning from scientific exploration to large-scale engineering deployment. To address the constraints of application, the training ecosystem should evolve from isolated algorithmic optimizations to systemic synergy. Future developments will synchronize across three axes: increasing data value density, establishing foundation model capabilities, and enabling continuous platform evolution, which can reinforce each other to define the next generation of robust and scalable autonomous systems.

A. Scenario Prioritization and Data Engine: From “Data Volume” to “Value Density”

Building on the value-driven governance defined in Section III, the primary engineering challenge remains the construction of automated value scoring systems that can operate at scale. While the necessity of “value density” is clear, implementing it requires overcoming significant bottlenecks in current data infrastructures.

First, collection processes rely on fixed routes or random strategies, with insufficient coverage of key scenarios and redundant samples occupying substantial resources. Additionally, annotation processes are inefficient, especially in multimodal data processing where spatiotemporal alignment problems remain unsolved, and lack of hierarchical management leads to high-cost annotation consumption on low-value samples. Finally, governance processes are fragmented, with incomplete version records and difficult traceability of additions and removals, thus limiting reproducibility of model iteration. To achieve continuous progress, systematic breakthroughs are urgently needed in the following aspects:

- **Construct scenario value systems.** Establish quantifiable and operational scenario value scoring systems that comprehensively consider risk, rarity, learnability, and coverage gaps to form unified priority lists. Implement hierarchical treatment and resource allocation based on thresholds, making collection and annotation focus on scenarios with the highest marginal returns.
- **Implement event collection and controllable synthesis.** In event-triggered collection, rely on vehicle-side shadow operation around trigger conditions such as emergency braking, sharp turns, and close encounters to prioritize recording “near-miss events” and capture critical decision samples from human driving. On the synthesis side, maintain semantic consistency and physical credibility, using programmatic generation and physics engines to construct counterfactual scenarios such as illegal cutting,

TABLE V
INDUSTRIAL PRACTICES IN AUTONOMOUS DRIVING

Company	Learning Mechanism	Testing Approach	Key Focus	Advantages	Challenges
Waymo	Continuous retraining with real-world validation	Simulation-first using Carcraft platform	High-fidelity, rule-based scenario generation for safety validation	High reproducibility and interpretability; rigorous safety validation	High scalability costs; limited adaptability to real-world conditions
Tesla	Continuous fleet learning with online retraining	Real-world testing through fleet data collection	Event-triggered data collection; large-scale fleet feedback; automatic data annotation	Scalable exposure to real-world conditions; rapid iteration of training	Weak reproducibility; regulatory challenges regarding data privacy and safety
Cruise	Incremental city-level training updates with controlled testing zones	Hybrid real-world and simulation testing	City-specific training and safety validation	Robust version control; effective safety monitoring within urban environments	Limited scalability across regions; expensive infrastructure for testing and validation
Baidu Apollo	Periodic retraining with continuous deployment in open-road pilots	Hybrid real-world and simulation testing	Government-integrated data governance and safety compliance	Strong regulatory compliance; consistent model reproducibility	Slower feedback adaptation; challenges in handling diverse driving environments
XPeng	Federated learning and asynchronous updates for cross-regional adaptation	Cloud-edge collaborative training and testing	Cross-region data federation for model adaptation and deployment	Efficient cross-domain updates; real-time adaptability to regional differences	High infrastructure costs; limited transparency in testing methodologies
Huawei ADS	Continuous learning with gray release and rollback mechanisms	Cloud-edge collaborative training and testing	Stable asynchronous updates and safety-focused rollback strategies	Reliable model adaptation with real-time updates; controllable risk management	Limited openness and transparency; high computational cost for training

complex yielding, and unconventional interactions. Meanwhile, synthetic data use should be strictly controlled, mainly using synthetic data for long-tail completion and robustness enhancement.

- **Optimize annotation strategies and processes.** Address multimodal spatiotemporal alignment, achieve hardware-triggered synchronization with controlled time errors, and improve geometric consistency through backend calibration optimization. Implement hierarchical annotation processes, where automatic annotation with sampling inspection to screen high-value samples is first used, followed by conducting fine annotation on selected samples, striving to improve effective annotation density and output quality without increasing overall costs.
- **Strengthen data version governance.** Establish version trees to manage data full lifecycle, clearly record sources, filtering rules, and applicable scope, clarifying the basis and timing for additions and removals. Map data versions one-to-one with model checkpoints, support historical effect traceability and scenario-level reproduction, improve problem localization and repair efficiency, and avoid ineffective repeated training and resource waste.

B. Universal Driving Foundation Models and Regional Adaptation: Strong Transfer, Weak Customization

E2E-AD is moving from scenario-specific optimization toward universal foundation models capable of operating across cities, road types, and driving cultures. Such models must provide consistent multimodal, long-sequence representations that align training objectives with planning and safety metrics, reducing the gap between offline accuracy and online reliability. They also need to reconcile efficient regional customization through lightweight adaptation, stable cross-domain transfer

without catastrophic forgetting, and interpretable decision-making for safety assurance.

However, current foundation models degrade sharply in new domains in practice. Differences in signage, rules, and driving habits often cause planning drift, while existing adaptation methods rely on costly retraining or extensive annotation and struggle with efficiency. Performance instability and catastrophic forgetting remain common, and the lack of clear criteria for when to retrain versus fine-tune further impedes deployment. Limited interpretability compounds these issues, making adaptation outcomes difficult to assess and verify. Future research needs technological breakthroughs in the following key directions:

- **Cross-domain knowledge transfer and unified representation learning.** Construct unified representation frameworks with cross-modal and cross-temporal consistency, enabling models to maintain stability across different cities and road conditions. Future work can combine contrastive learning, domain adaptation, and causal representation learning to enhance model transfer capabilities under changes in road rules, signage styles, and driving habits, thereby reducing needs for re-annotation and training.
- **Lightweight regional adaptation and parameter-efficient updates.** Achieve rapid updates for new cities or road styles through lightweight adaptation layers or prompt tuning methods. Research how to use small-scale incremental data for parameter low-rank updates or sparse adaptation while ensuring performance while reducing computational and storage overhead.
- **Stability and forgetting suppression mechanisms.** Introduce replay learning, knowledge distillation, and regularization methods in cross-domain updates to effectively

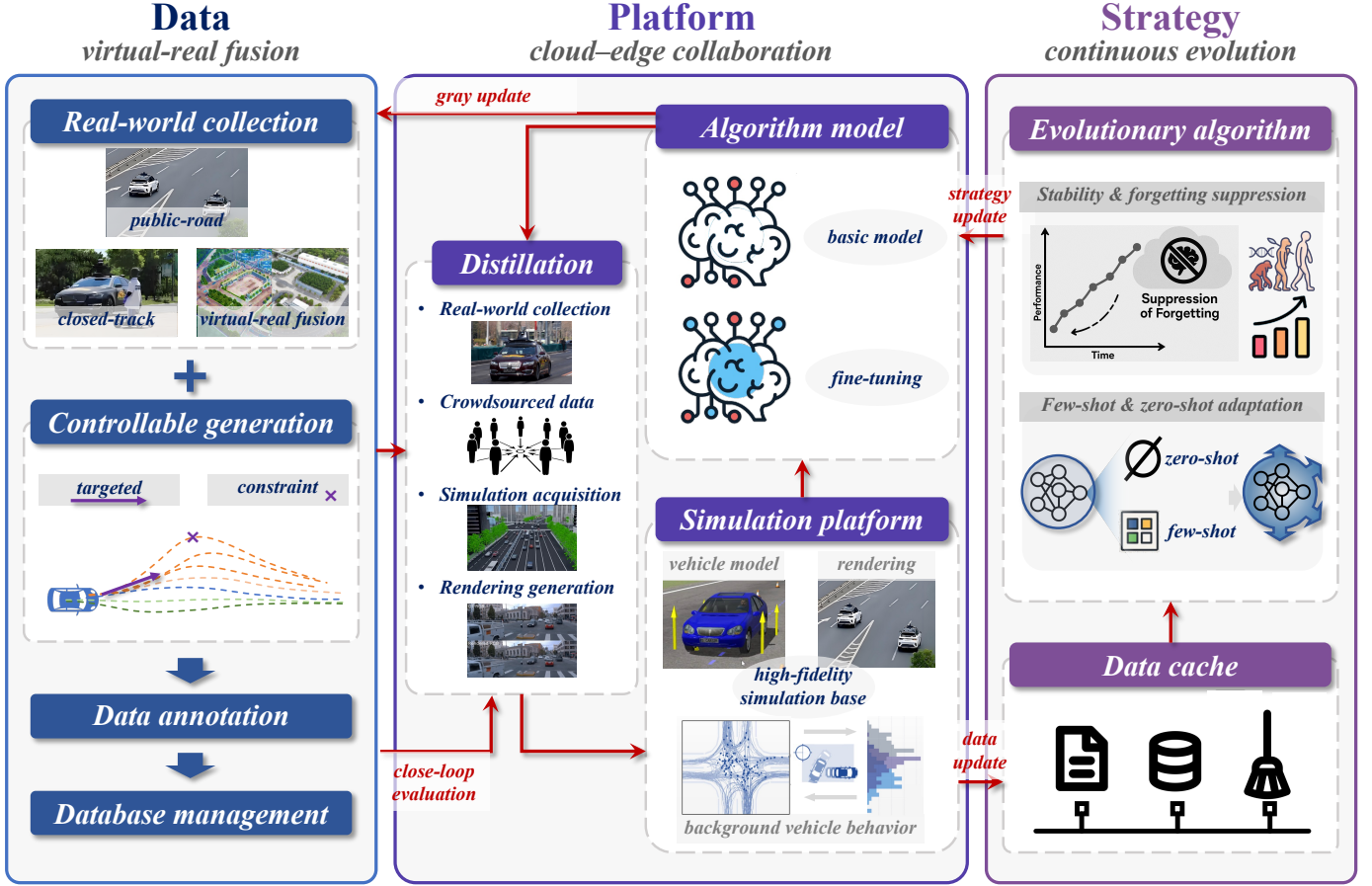


Fig. 6. Schematic diagram of the Data-Strategy-Platform coupled training system, demonstrating the closed-loop cycle of continuous evolution. **(Left) Data Virtual-Real Fusion:** Real-world failures trigger controllable generation to create targeted training sets. **(Middle) Cloud-Edge Collaboration:** The Platform layer manages the distillation of heavy models into deployable units, facilitating gray updates. **(Right) Strategy Evolution:** The Strategy layer employs evolutionary algorithms to adapt to new scenarios while suppressing catastrophic forgetting. The red arrows indicate the critical feedback loop from online deployment back to offline optimization.

suppress forgetting and reduce performance fluctuations. Meanwhile, develop dynamic judgment mechanisms and establish quantitative metrics to determine when comprehensive retraining is needed versus when only small-scale parameter adjustments are sufficient, thereby achieving efficient iteration.

C. Continuous Learning and Training Operations: Closed-Loop Training Solutions with Training-Testing Integration

To enhance E2E-AD as they move from closed sites to urban environments, it is essential to address the increasing complexity of scenarios, environmental variability, and strict safety requirements. A critical challenge arises from the persistent separation of training and testing processes: traditional methods rely on static datasets and one-time evaluations, which cannot capture the frequency and diversity of real-world changes. The resulting disconnect creates a significant gap between models that perform well in offline tests but fail to generalize effectively to unpredictable, real-world situations.

Addressing these limitations requires training-testing integration shown in Fig. 6. As illustrated in the **left panel**, the process begins with *data virtual-real fusion*, where real-

world corner cases are augmented by controllable generation to ensure high value density. These inputs feed into the **central Platform layer**, which manages the *distillation* of large foundation models into vehicle-embedded versions. Finally, the **right panel** depicts *strategy continuous evolution*, where techniques such as evolutionary algorithms and forgetting suppression are applied to ensure that updates improve performance without degrading previously learned capabilities. However, constructing such a system requires solving the following core tasks:

- **Construct unified data-model-evaluation lineage systems.** Currently, training, testing, and deployment often belong to different platforms, lacking unified metadata structures and process records, making model performance difficult to trace and version comparisons difficult to reproduce. Future work needs to establish consistent lineage management mechanisms among training samples, model versions, and evaluation results, clarifying data inputs, optimization configurations, and validation outputs that each update depends on, supporting scenario-level reproduction verification, thereby providing clear technical basis and audit traceability for iteration paths.

- **Develop model evaluation systems based on closed-loop metrics.** Traditional evaluations mostly focus on accuracy metrics, making it difficult to cover strategy reasonableness and safety redundancy in end-to-end decision chains. Future work needs to construct task-level closed-loop evaluation metric systems around “systematic stability, decision reasonableness, risk sensitivity,” combining high-risk case replay, counterfactual perturbation verification, and simulation result comparison to achieve systematic evaluation of model behavior stability and robustness, enabling training results to truly map to online performance.
- **Establish reproducible minimum closed-loop mechanisms for training-validation-deployment.** Minimum training closed-loop paths centered on problem-driven, metric-constrained, and resource-sensitive approaches should be constructed. Starting from failure sample triggers, determining problem attribution through scenario replay and label verification, executing targeted retraining with minimal data increments, conducting performance confirmation through standardized validation sets, and finally supporting phased deployment and automatic rollback strategies. This mechanism needs three major characteristics: controllable processes, controllable resources, and controllable benefits, serving as underlying capabilities supporting large-scale evolution of autonomous driving.

VII. CONCLUSION

Training constitutes the cornerstone of E2E-AD, determining both performance boundaries and the trajectory of capability evolution. While unified modeling promises advantages over modular pipelines, the training process remains constrained by long-tail data scarcity, unstable optimization, and platform bottlenecks. The central insight of our Data-Strategy-Platform analysis is that these challenges are not isolated hurdles but systemic dependencies that require a synchronized evolution of the training ecosystem. Future progress, therefore, depends less on individual breakthroughs in any single layer and more on the synergy across all three. It is this collaborative optimization of data quality, strategic alignment, and platform scalability that will unlock the reliability required for large-scale, sustained real-world deployment.

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